4.4 Relating Key Features of Lines to Real-World Contexts

Lesson Introduction

Benchmark	MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function and determine and interpret its key features.	Learning Targets	• I can locate the key features of a linear function using graphs, tables, equations, and written descriptions. • I can determine if the domain and range are viable for the context. • I can interpret the slope and y-intercept based on the context and determine if the interpretations are viable.

	Building On	Working Toward
Benchmark Coherence	MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.	MA.912.F.1.7 Compare key features of two functions each represented algebraically, graphically, in tables or written descriptions.

Essential Questions	• What can key features tell me about a linear function? • How can domains be mathematically appropriate, yet not viable for the context? • How can the slope and y-intercept be interpreted in a given context?
Lesson Narrative	In the lesson, students will determine the viability of key features within a given context. In the prior two lessons, students have determined key features of functions and are moving in this lesson from identifying to interpreting. Students will then determine if the values are viable based on a given real-world context. This lesson will provide students with a gallery walk activity for student practice.

Component Summary	4.4.1 Warm-Up (~10 min.)	Matching equations, tables, and graphs using key features *(SE p. 179)*	4.4.3 Activity (15 min.)	Students complete a gallery walk to determine if a key feature is viable *(SE p. 181)*
	4.4.2 Collaboration (15 min.)	Students collaborate on interpreting key features and determining viability *(SE p. 179)*	4.4.4 Wrap-Up (~10 min.)	Students determine if the given key feature is viable and explain why or why not *(SE p. 183)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Domain • Viable <u>Additional Terms:</u> • Intercepts • Key Feature	• Sticky notes • Chart paper or poster board alternative • Markers	• 4.4.3 Gallery Walk requires logistical preparation. Consider who you will group together (2-4 per group) and to whom you will assign each context. • Read through each context and determine if there are any potential pairings based on student abilities.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle to determine the key features of a graph without being prompted for specific features.
Students may forget to determine the viability of an answer in context without a prompt.
Students may have trouble determining if a value is viable as the context changes from one problem to the next.</td><td>Consider what information or data may be used for grouping students for 4.4.3 Activity based on observations from Lessons 2 and 3.
The gallery walk uses materials like chart paper and sticky notes that should be prepared prior to delivering the lesson.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
4.4.3 Activity provides an opportunity for students to present their work and then receive written feedback from their peers through the gallery walk process. Students will post their initial work and then visit each group to provide questions and feedback.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
4.4.3 Activity is a gallery walk activity that provides students with the opportunity to communicate the viability of key features to others. Additionally, students will view the explanations of their peers and provide questions and feedback.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">4.4.3 Activity highlights various learning styles. Visual learners will benefit from the visual of the completed posters from the gallery walk. Auditory learners will benefit from the collaborative conversation. Kinesthetic learners can point, follow, or trace linear functions on a larger scale, all while using movement to their advantage.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

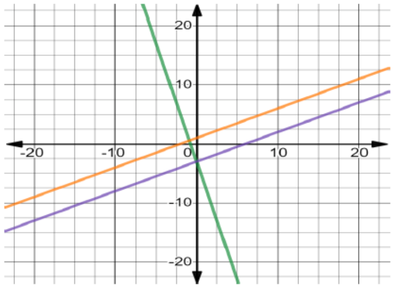
4.4.1 Warm-Up: Matching

Component Overview: *Students apply their knowledge of key features of graphs to match tables, equations, and graphs of linear functions.*

4.4.1 Warm-Up: Matching

1. Match the equations, tables, and graphs noting any key features you use to determine the matches.

	Equations
A	$4x + y = -3$
B	$y = \frac{1}{2}x - 3$
C	$y - 4 = \frac{1}{2}(x - 6)$



	Tables						
1	x	-3	-2	-1	0	1	2
	y	-4.5	-4	-3.5	-3	-2.5	-2
2	x	-3	-2	-1	0	1	2
	y	9	5	1	-3	-7	-11
3	x	-3	-2	-1	0	1	2
	y	-0.5	0	0.5	1	1.5	2

Equation	Table	Color of Graph	Key Features I Used
A	2	*Green*	*Answers will vary. Slope and y-intercept.*
B	1	*Purple*	*Answers will vary. x-intercept and y-intercept.*
C	3	*Orange*	*Answers will vary. Slope and a point.*

This activity should be completed independently as progress monitoring for the three-part lesson series. Additionally, keep in mind that the following activity, 4.4.2 Collaboration, explicitly provides an additional entry point for students to make sense of their answers from the Warm-Up. If time permits, consider allowing students to verify their answers with a partner.

4.4.2 Collaboration: Key Features From a Graph

Component Overview: *Students will work with a partner to discuss their findings and methods from the Warm-Up and begin interpreting key features and viability within a given context.*

4.4.2 Collaboration: Key Features from a Graph

1. With your partner, discuss which key features they used for each graph and why they used those key features to determine the equation and table that matched the graph. Record your summary of their key features and reason. You are the only person who should write in the middle column of the table. Have your partner initial next to your summary stating that your summary was correct.

Graph	My Summary of My Partner's Key Features	Initials
Orange	*Answers will vary. My partner chose the x- and y-intercepts for their key features because they were able to quickly solve for each when in standard form. My partner chose the slope and y-intercept because the y-intercept was easy to find and the slope made it easy to see the line progress.*	
Green	*Answers will vary. My partner chose the x- and y-intercepts for their key features because they were able to quickly solve for each when in standard form. My partner chose the slope and y-intercept because the y-intercept was easy to find and e progress.*	
Purple	*Answers will vary. My partner chose the x- and y-intercepts for their key features because they were able to quickly solve for each when in standard form. My partner chose the slope and y-intercept because the y-intercept was easy to find and the slope made it easy to see the line progress.*	

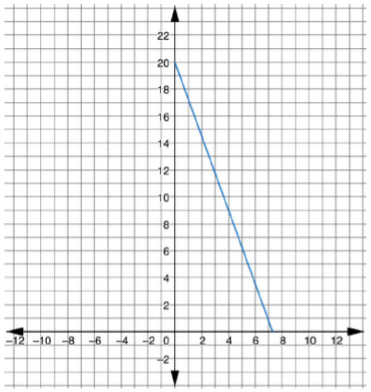
Students' conversations should utilize academic vocabulary such as x-intercepts, y-intercepts, key features, domain, and/or range. Consider providing scaffolds like a Word Wall or Table Cards to support mathematical discourse.

Listen for equity of voice in the discussion to gather formative data for student understanding and potential misconceptions about key features. Is one student talking more than the other? Set timers for the whole group and specifically identify turns when discussing.

Consider a unique and fun game of "Math Taboo" when discussing responses. One student has a Taboo card of words that students use instead of math vocabulary. That student oversees monitoring the math talk and taking away Taboo points whenever one student says a Taboo word. This is an engaging and competitive way for students to build math vocabulary in their math talk.

4.4.2 Collaboration: Key Features From a Graph (continued)

2. Bella is using a gift card to buy coffee at a coffee shop. The initial value of the gift card is $\$20$. The function $A(c) = 20 - 2.75c$ represents the amount of money, A, in dollars, that is still left on the gift card after purchasing c cups of coffee.



c	A
0	20
1	17.25
2	14.5
3	11.75
4	9
5	6.25
6	3.5
7	0.75
$7.\overline{27}$	0

Work with your partner to identify the key features of the function $A(c) = 20 - 2.75c$.

Domain	**Range**	**As x-values increase, y- values …**
$0 \leq x \leq 7.\overline{27}$	$0 \leq y \leq 20$	Decrease
Slope	**x-intercept**	**y-intercept**
-2.75	$7.\overline{27}$	20

The ***domain*** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

The definition of the domain states, "The domain may vary depending on the context." The domain of the function $A(c) = 20 - 2.75c$ should only be whole numbers 7 or less.

The conversation can provide qualitative data to determine the degree to which students understand identifying key features of graphs.

It may also provide evidence of student understanding of graphs, tables, and equations for a future opportunity for differentiation.

Add a challenge by asking students to write the domain and range values using set-builder notation.

What does it mean that the domain "may vary depending on the context" in this definition?

4.4.2 Collaboration: Key Features From a Graph (continued)

3. Work with your partner to write an explanation for why the domain should only be whole numbers 7 or less. Then, if necessary, fix your domain for $A(c) = 20 - 2.75c$ in the previous problem.

   Answers will vary. This is a real-world scenario. The domain should only be whole numbers to best match the scenario. In this example, the x-axis represents cups of coffee, which should be limited to whole numbers.

4. Complete the interpretation of the slope of the function.

   For every ○ 1 cup ○ 2.75 cups of coffee purchased, the amount of money left on the gift card ○ decreases ○ increases by ○ $\$1.00$. ○ $\$2.75$.

5. Work with your partner to explain what the y-intercept indicates about the gift card.

   The original amount of money placed on the gift card.

A key feature is **viable** if the value(s) are feasible for the real-world context.

6. The x-intercept for the function $A(c) = 20 - 2.75c$ is not viable for the context. Explain why the x-intercept does not make sense for the context.

   Answers will vary. Student answers should reflect that a person cannot buy a partial cup of coffee.

Collaboration in question 3 should have students reasoning through why the domain in the context should only be whole numbers. Discussions should reference specific language about the cups of coffee and/or gift card.

Language should be specific in rationale, not overly general.

What is another situation where the domain should only be whole numbers? Why?

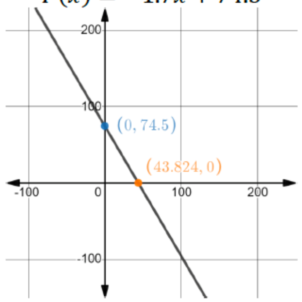
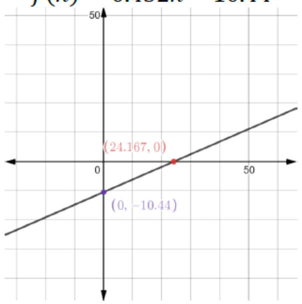
Is there another real-world scenario where this answer would be viable?

4.4.3 Activity: Gallery Walk

Component Overview: *Students consider four contexts to determine the key features based on the context.*

4.4.3 Activity: Gallery Walk

There are four contexts shown. Use the description of the context and representations of the function to determine if each key feature on your recording sheet is viable in context.

Context	Description	Representations
A	The percentage, P, of U.S. companies who performed drug tests on employees or job applicants for x, the year since 1998, for years 1998-2003	$P(x) = -1.7x + 74.5$ 
B	The length of a woman's femur, f, in inches, as a function of her height, h, in inches	$f(h) = 0.432h - 10.44$ 

For each context, students should identify each key feature then determine if each key feature is viable.

- *Option 1: Each context with description and representation is posted on a different wall of the room for students to explore in pairs.*
- *Option 2 (Card Sort): Each description and representation are separate cards, so each group of students has eight cards shuffled in random order. Groups first match each description to a representation then identify the key feature and its viability.*
- *Option 3: Divide the class into groups of 2-4 students. Give each group a set of key features to determine and discuss. This can be all the key features for a certain context. or different key features for each context.*

Sample Questions for Context A:

- *Is there a difference if the answer is a complete year or a partial year? Give an example.*
- *Does 43.824 as 43 years after 1998 make sense in this context? Why or why not?*

Sample Questions for Context B:

- *Does $(24.167,0)$ as the x-intercept makes sense in this context? What would that mean for the femur?*
- *Discuss reasonable heights for women when discussing the domain. When would this context not make sense? For children, for example?*

4.4.3 Activity: Gallery Walk (continued)

C	Sarah's parents give her a $\$100$ allowance at the beginning of each month. Sarah spends her allowance on comic books. The amount of allowance Sarah has remaining, y, in dollars, is a function of the number of comic books, x, that she buys for $\$5$ each.	$y = -5x + 100$ (table below)
D	A construction company charges $\$8$ per hour for debris removal plus a one-time fee of $\$150$ for the use of a trash dumpster. The total fee, y, in dollars, is a function of the number of hours, x, the company takes to remove the debris.	$8x - y = -150$ (table below)

$y = -5x + 100$

Number of Comic Books Purchased	Allowance Remaining
2	$\$90$
9	$\$55$
13	$\$35$
20	$\$0$
24	$-\$25$

$8x - y = -150$

Number of Hours	Fee (\$)
-10	$\$70$
-4	$\$118$
0	$\$150$
5	$\$190$
24	$\$342$

Gallery Walk Recording Sheet

Context	Domain	Range	x-int.	y-int.
A	Answers will vary	Answers will vary	$(43.824, 0)$ not viable	$(0, 74.5)$ viable
B	Answers will vary	Answers will vary	$(24.167, 0)$ not viable	$(0, -10.44)$ not viable
C	Answers will vary	Answers will vary	$(20, 0)$ viable	$(0, 100)$ viable
D	Answers will vary	Answers will vary	$(-18.75, 0)$ viable	$(0, 150)$ viable

Sample Questions for Context C:

- *Can Sarah purchase 24 comic books? Why or why not?*
- *Is 24 part of a viable domain? Why or why not? What is the greatest number possible that would still be viable?*

Sample Questions for Context D:
What could the domain tell us about the viability of our solutions? Is this always true? Give an example.

Student answers may vary for the domain and range for each context. The viability depends on the domain and range that each student identifies. Using the questions provided, further explore the values and the viability that students identify.

Context An Example (Domain):

- *Student 1: $0 \le x \le 43.824$ (Not Viable)*
- *Student 2: $0 \le x \le 5$ (Viable)*

Student 1 may respond with "Not Viable" since the context only specifies years 1998-2003, while Student 2 may respond with "Viable" since $0 \le x \le 5$ explicitly identifies the context of years 1998-2003 only.

Both students are correct. Instead of focusing on the "correct" answer, discussion should be centered on determining viability in a variety of real-world contexts.

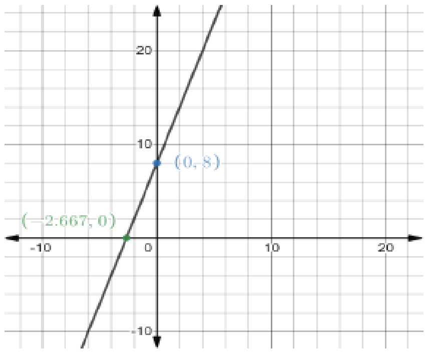
4.4.4 Wrap-Up: Ticket Out the Door

Component Overview: *Students will complete a ticket out the door that provides students with a formative assessment on determining viability given a key feature and explaining why the key feature is or is not viable.*

4.4.4 Wrap-Up: Ticket Out the Door

Antonella places x marbles into an empty container. Each marble has the same weight.

The weight, W, in ounces, of the container and the marbles can be found using the equation $W(x) = 3x + 8$.



1. Li Qiang completed the table of four key features. Determine if the key features given by Li Qiang are viable. If the given key feature is not viable for the context, explain why it is NOT viable.

	Domain	Range	x-intercept	y-intercept
Key Feature	$x \geq 0$	$y \geq 0$	$\left(-\frac{8}{3}, 0\right)$	$(0, 8)$
Viable? Yes/No	*Yes*	*No*	*No*	*Yes*
Explanation	*The number of marbles she has in her bag cannot be negative.*	*8 is the weight of the empty container, so y cannot be any value less than 8.*	*The x-intercept would imply that there are a negative number of marbles with no weight.*	*This y-intercept would represent when there are 0 marbles. This means the container weighs 8 ounces.*

The data from this Wrap-Up will allow you to differentiate your future instruction or provide remediation for students on this topic.